

Investigating Search Engines Terms Leakage

Mateo Ormeno

Pontifical Catholic University of Chile

Ha Dao

Max Planck Institute for Informatics

Kensuke Fukuda

National Institute of Informatics

ABSTRACT

This paper delves into the growing concerns about user privacy on the Internet, emphasizing the lack of transparency in web tracking. We study the phenomenon of search engines' keyword leakage – instances where search terms inputted by users are unintentionally and unknowingly transmitted to third parties. We conduct automated experiments on the top 15 popular search engines and repeat them in three distinct locations: Japan, Chile, and the Czech Republic. Preliminary findings reveal that six of the 15 examined search engines exhibited leakage, showing slight differences across regions. These results emphasize the need for further analysis to characterize this privacy leakage.

CCS CONCEPTS

• **Information systems** → *Web crawling*; • **Security and privacy** → *Privacy protections*.

ACM Reference Format:

Mateo Ormeno, Ha Dao, and Kensuke Fukuda. 2023. Investigating Search Engines Terms Leakage. In *Proceedings of the CoNEXT Student Workshop 2023 (CoNEXT-SW '23)*, December 8, 2023, Paris, France. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3630202.3630236>

1 INTRODUCTION

Search engines have become an efficient and reliable tool for exploring the Internet. They help users find web pages that match their search queries. However, there are increasing concerns about online privacy due to potential data leakage to third parties without user knowledge [2].

Studies have addressed the growing sophistication of web tracking techniques and privacy issues related to information leakage. Various aspects of online tracking can lead to privacy violations, from device-specific data to Personal Identifiable Information (PII) [5]. Previous research has identified PII leaks in registration forms [4]. A recent study also revealed the extent of third-party access to search terms used in internal site search boxes [7] through HTTP requests. However, there are no empirical records of studies on search engine users' privacy integrity, indicating a potential risk of users' search terms reaching non-desired third parties. As search engine users' searches may include PII, sensitive data, or even slight "hints" that could lead to identifying them – something that search engines are known to have used in the past [1] –, the potential risk is significant when third parties gain access to users' search engine

identities. They could use this information for malicious purposes such as blackmailing, impersonation, and other harmful activities.

Our study focuses on the potential exposure of users' search terms to third parties when using popular search engines. We conduct automated experiments using web crawlers and analyze their browser activity data. Our early-stage findings show that six of the 15 search engines tested exhibit leakage, a severe concern for users. Also, we see a similar leakage behavior by repeating the experiments in different countries with varying privacy regulations, including Japan, Chile, and the Czech Republic. However, the extent varies slightly across them. These results support search engine users' concerns about their exposure and raise awareness about online privacy.

2 METHODOLOGY

2.1 Data collection

Our first task is to select 15 search engines to study their leakage behavior across various parameters such as the number of users, country of origin, and privacy focus. To cover the most significant number of users, we choose 12 of the most popular search engines according to their worldwide market share¹: *google*, *bing*, *yahoo*, *yandex*, *duckduckgo*, *baidu*, *sogou*, *naver*, *ecosia*, *seznam*, *qwant*, and *startpage*. This selection also includes search engines from different regions – the USA, Europe, and Asia – with varying levels of privacy focus. We also evaluate three other search engines: *swisscows* for its novelty (2014), *aol* for its relevance to our study [1], and *ask* for its format as Q&A to provide the searching service.

For each search engine, we obtain its *search URL* directly from a web browser by manually performing an arbitrary search query on their main page, so we get the most accurate and updated version of their URLs. Then, for each *search URL* collected, we can replace the desired search term substring in the corresponding placeholder as a query parameter, creating a list of visitable URLs for our web crawlers. We use OpenWPM [6], a Python framework for web privacy measurements that provides web crawling capabilities over Firefox browsers that store the network data in SQLite databases. We crawl the 15 *search URLs* replacing the search term "*JELLY-BEANS*"², each in a separate fresh browser. We run ten web crawls for each search engine in separate, fresh browsers without logging in to any service (e.g., Google account).

To represent a wider variety and cover more users, we repeated the experiment from three different locations using NordVPN: Tokyo, Japan; Santiago, Chile; and Prague, Czech Republic. By doing so, we have gained a more realistic understanding of the privacy situation for users in three different continents – Asia, South America, and Europe – including the case of a country under the General Data Protection Regulation in Europe³ (Czech Republic). This law requires organizations in the EU to adhere to hundreds of pages

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CoNEXT-SW '23, December 8, 2023, Paris, France

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0452-9/23/12

<https://doi.org/10.1145/3630202.3630236>

¹<https://gs.statcounter.com/search-engine-market-share#monthly-202204-202304>

²This search term string was picked for simplicity in the experiment, following Ref. [7].

³<https://gdpr-info.eu/>

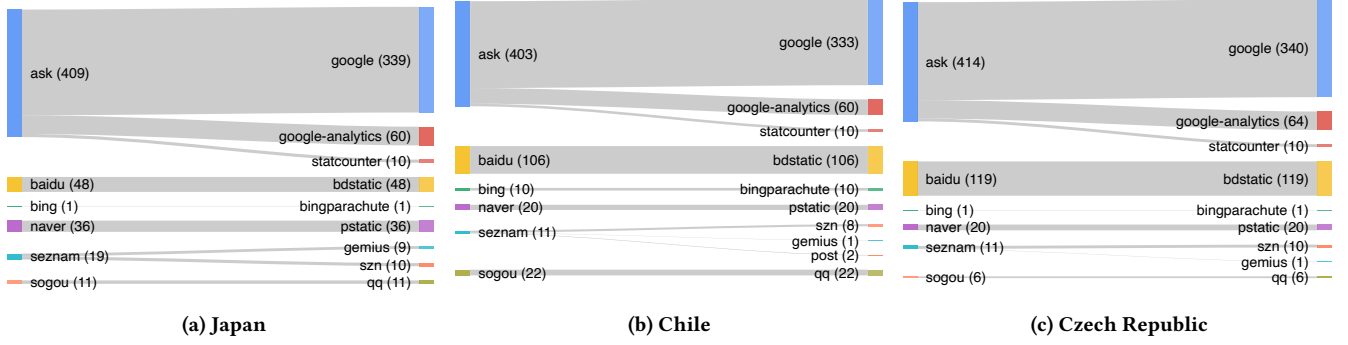


Figure 1: Breakdown of third-party domains involved in term leakage among six search engines. The amount of data captured is based on ten crawls.

Table 1: Summary of crawled data in 15 search engines (August 2023).

Metric	Japan	Chile	Czech Republic
HTTP requests	13,345	10,778	12,157
Javascripts	22,735	23,700	16,216
Cookies	2,433	1,952	1,885

of requirements on strict user data privacy and security, which should imply better protection against leakages. The Summary of our dataset is shown in Table 1.

2.2 Leakage detection

Our study defines a *third-party* as the domain of a URL with a second-level domain different from the domain of the search engine used. We consider it a *third-party keyword leakage* when the keyword (i.e., the search term inputted in the search engine) appears in any accessible piece of browsing activity data sent to a *third-party* captured during the search. Our approach involves searching for the search term within the relevant fields of the collected browser activity data to third parties to identify the *third-party keyword leakages* in a similar approach as Ref [5]. We conduct this lookup using standard and lowercase versions of the keyword, utilizing 25 encodings and hash functions for each variation in case the keyword was encoded. We focus our attention on three specific elements: (i) *HTTP Requests*: where we check the *request_url*, *post_body*, *referrer* and *headers*. (ii) *Javascripts*: where we check only the *value* field. (iii) *HTTP Cookies*: where we check the *name*, the *path* and the *value* fields.

3 PRELIMINARY RESULTS

Figure 1 shows *third-party keyword leakages* on six out of 15 search engines by location. The search engines' names appear on the left column, and the third party's second-level domains are on the right column. The size of the connection between both columns' elements represents the amount of captured browsing data found to have a leakage of the keyword "JELLYBEANS" in any of the encodings. The exact amount appears inside brackets for each element. Overall, a steady number of leaks is detected for both *http_req_leaks* and *javascripts*. In particular, leakage is found in the same six search

engines and third parties across the three locations. Ask is the most prominent leaker for both HTTP Requests and Javascript for every location, leaking towards *google*, *google-analytics*, and *statcounter*. The ranking of the remaining search engines' leakages varies by location. Still, the numbers remain close, between 10-20 leaks per search engine, except for *bing*, which leaks ten times to *bingparachute* in the Czech Republic, while in Tokyo and Santiago, it only does once. *Baidu* (Chinese engine) has between 50-120 *Javascript* leaks from location to location, and it only leaks to the third party *bdstatic*. The rest of the leaking search engines are Naver (South Korea), Seznam (Czech Republic), and Sogou (China).

4 CONCLUSION

In this paper, we propose an experiment that delves into the privacy of search engine users, revealing term leakages across various popular platforms. Our findings emphasize reinforced data protection measures to safeguard user privacy. For future work, it is encouraged to expand our experiments by exploring a broader range of configurations (i.e., HTTPS to measure the impact of encryption on the tool's effectiveness) and locations (i.e., comparing more countries under the GDPR). Also, we aim to explore potential Online Behavioral Advertising [3] as an outcome of keyword leakage.

5 ACKNOWLEDGEMENTS

The authors thank the NII internship program.

REFERENCES

- [1] 2006. *A Face Is Exposed for AOL Searcher No. 4417749*. Retrieved September 18, 2023 from <https://www.nytimes.com/2006/08/09/technology/09aol.html>
- [2] Gunes Acar, Christian Eubank, Steven Englehardt, Marc Juarez, Arvind Narayanan, and Claudia Diaz. 2014. The web never forgets: Persistent tracking mechanisms in the wild. In *Proceedings of ACM SAC*. 674–689.
- [3] Juan Miguel Carrascosa, Jakub Mikians, Ruben Cuevas, Vijay Erramilli, and Nikolaos Laoutaris. 2015. I always feel like somebody's watching me: measuring online behavioural advertising. In *Proceedings on CoNEXT*. 1–13.
- [4] Manolis Chatzimpyros, Konstantinos Solomos, and Sotiris Ioannidis. 2019. You shall not register! detecting privacy leaks across registration forms. In *Computer Security*. Springer, 91–104.
- [5] Ha Dao and Kensuke Fukuda. 2021. Alternative to third-party cookies: investigating persistent PII leakage-based web tracking. In *Proceedings of the CoNEXT*. 223–229.
- [6] Steven Englehardt and Arvind Narayanan. 2016. Online tracking: A 1-million-site measurement and analysis. In *Proceedings of ACM CCS*. 1388–1401.
- [7] Daniel Kats, David Luz Silva, and Johann Roturier. 2022. Who knows I like Jelly Beans? An investigation into search privacy. *Proceedings on PETS 2022*, 2 (2022), 426–446.